

Reviving Anomalies

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Abstract

Prior literature finds that most anomalies fail to deliver significant returns, especially when considering implementation costs. We develop a framework to revive anomalies by incorporating expected returns and transaction costs into portfolio formation. Our approach identifies implementable anomalies across different fund sizes by (i) using machine learning return forecasts and (ii) accounting for scale-dependent price impacts. A 1/N investment strategy in implementable anomalies yields out-of-sample Sharpe ratios between 0.64 and 1.01 with highly significant alphas across all fund sizes. We further document that mutual funds tilt their holdings toward implementable anomalies and that such tilts predict higher subsequent fund returns.

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JEL classification: G10, G11, G12

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1. Introduction

In this paper, we revisit the recent performance of well-known return anomalies, like size, value, and momentum, which serve as the building block of many mutual funds and other investment vehicles. Academic evidence suggests that most anomalies have failed to deliver significant returns over the past twenty years (Green, Hand, and Zhang, 2017), especially when considering their implementation costs that can be prohibitive with high turnover rates (Chen and Velikov, 2023). Green et al. (2017) point to multiple reasons for the decay in anomaly performance, including a decrease in the level of mispricing due to regulatory changes. While we do not pin down a specific cause, we acknowledge that the decay coincides with the advent of more accurate methods to measure expected returns.

We develop a framework to address the declining performance of simple long-short anomalies by double-sorting portfolios based on the respective anomaly and expected returns net of transaction costs. To estimate expected returns, we follow Gu, Kelly, and Xiu (2020) and employ machine learning return forecasts based on the 153 stock characteristics documented in Jensen, Kelly, and Pedersen (2023). To quantify expected transaction costs, we differentiate between fund sizes and incorporate a scale-dependent price impact model, following Jensen, Kelly, Malamud, and Pedersen (2025) and Frazzini, Israel, and Moskowitz (2018). One major advantage of our framework is that it preserves the interpretability of standard factor portfolios while incorporating modern machine learning methods. Specifically, our approach serves as an overlay that identifies the most promising investment opportunities for each return factor across different fund sizes.

We start by showing that only a small number of anomalies produced significantly positive returns net of price impact costs for funds with varying assets under management (AUM) in an ex post analysis starting in 2004. The four fund sizes chosen (micro, small, medium, and large) are inspired by the distribution of fund sizes in the CRSP Mutual Fund Database, roughly reflecting the 20th, 40th, 60th, and 80th percentiles. The fund sizes are measured in 2023 and discounted back in time with the market return, following Jensen et al. (2025). For small funds managing 200mn USD, 10 of the 153 characteristic-based anomalies documented in Jensen et al. (2023) generated significantly positive net returns of 8% on average, with an average Sharpe ratio of 0.60. In hindsight, micro funds managing 40mn USD would have had 17 and medium-sized funds managing 500mn USD 3 anomalies with significant returns at their disposal. For large funds, managing assets in excess of 1.6bn USD, only a single anomaly generated significantly positive net returns

over the past two decades.

We then show that incorporating next-month expected returns of the constituents into the formation of the anomaly portfolios substantially increases the number of anomalies with a significantly positive net return over the past 20 years. Our expected return measure uses stock characteristics as conditioning information and follows the neural network approach developed in [Gu et al. \(2020\)](#), which has shown great success in this domain across asset classes ([Bali, Beckmeyer, Moerke, and Weigert, 2023](#)). Specifically, we perform conditional quintile sorts, first by the anomaly portfolio’s conditioning characteristic, and then by the stocks’ expected returns. In the long leg of each anomaly, we retain only those stocks in the highest expected return quintile, whereas we retain only the stocks in the lowest expected return quintile for the short leg.¹ For small funds, incorporating expected returns increases the number of significant anomalies to 73 with net returns of 18% and a Sharpe ratio of 0.70 on average. Micro funds would have had 126 and medium-sized funds 9 significant anomalies, if the selection of applicable stocks also considers their expected next-month return. For the largest funds, not a single anomaly’s returns are significantly positive, even after this treatment.

To account for implementation costs, we estimate expected transaction costs using a scale-dependent price impact model, following [Jensen et al. \(2025\)](#) and [Frazzini et al. \(2018\)](#). We then adjust expected returns for the estimated price impact within our conditional quintile sorts. Once the portfolio formation considers expected returns and transaction costs of the included stocks, we find that small funds have access to 101 significant anomalies with net returns of 14% and a Sharpe ratio of 0.71 on average. Compared to the previous double sort based solely on expected returns, the number of significant anomalies does not change for micro funds and stays at 126. However, the joint consideration of expected returns and expected transaction costs yields a set of 81 significant anomalies for medium-sized funds and 41 for large funds. These findings indicate that our framework is effective across all fund sizes. Micro and small funds particularly benefit from our expected return extension, whereas expected transaction costs are most critical for medium and large funds due to their substantial price impact. In terms of anomaly themes, our approach exhibits strong performance for quality- and accrual-based strategies. Notably, we are unable to revive any size-based anomalies for fund sizes ranging from small to large, highlighting their dependence on illiquid, small-cap

¹We show that this approach is vastly different from moving to finer quantiles, such as deciles, or ventiles.

stocks.

Naturally, this historical analysis is silent on identifying the set of anomalies that are *expected* to generate positive returns in the future and therefore does not directly inform the construction of a systematic investment strategy. We therefore investigate the extent to which this ex post evidence translates into improved out-of-sample investment performance for hypothetical funds with varying AUM. In an ex ante analysis, we evaluate how many of the 153 characteristic-based anomalies are expected to yield positive returns net of transaction costs for a given fund size, which we define as *implementable* anomalies. A simple 1/N investment strategy based on implementable anomalies achieves economically meaningful net returns across micro, small, medium-sized, and large funds. For instance, a small fund generates average returns net of transaction costs of 12.0% over our 20-year out-of-sample period, with a Sharpe ratio of 0.90 and a CAPM α of 15.5% per year. Even a large fund achieves an average net return of 5.4% with a Sharpe ratio of 0.64 over the same period, alongside a highly significant CAPM α of 7.7%, indicating substantial opportunities relative to a market-only investment.

To assess the economic significance of our results, we combine information on which anomalies are expected to be implementable at a given fund's AUM with the fund's portfolio holdings from the CRSP mutual fund database at the fund-month level. We then introduce a novel measure, *anomaly alignment*, which captures the average overlap between a mutual fund's holdings and our anomaly portfolios adjusted for expected returns and transaction costs, for both implementable and non-implementable anomalies. Economically, a higher level of anomaly alignment indicates that a fund more closely tracks our augmented anomaly portfolios on average. We show that mutual funds across all fund sizes tilt their holdings toward implementable anomalies and away from non-implementable anomalies, suggesting that they indirectly incorporate expectations about returns and transaction costs into portfolio formation.

In a regression setting, we show that mutual funds whose holdings are more closely aligned with the set of implementable anomalies at their respective AUM levels generate substantially more value for investors, as reflected in significantly higher fund returns net of expenses. Cross-sectional regressions indicate that an interquartile range (IQR) increase in the average anomaly alignment of implementable anomalies raises the average monthly return by 0.71 percentage points, corresponding to an annual return differential of 8.48 percentage points. Panel regressions further show that a one IQR increase in current anomaly alignment of implementable anomalies increases fund returns in the subsequent

month by 0.50 percentage points, which translates into an annualized difference of 6.02 percentage points. Notably, variation in anomaly alignment among non-implementable anomalies has no significant effect on fund returns in either cross-sectional or panel regressions. This finding provides empirical support for our prediction that mutual funds can exploit only those return anomalies that are implementable given their fund size.

Interestingly, the outperformance in net returns is not offset by higher management fees, suggesting that, on average, mutual funds do not charge investors for holding portfolios that are more closely aligned with implementable return anomalies. Neither cross-sectional nor panel regressions indicate a significant relationship between anomaly alignment of implementable anomalies and management fees. In contrast, mutual funds whose holdings are more closely aligned with non-implementable anomalies charge significantly lower management fees, suggesting that investors recognize the distinction between implementable and non-implementable anomalies for a given mutual fund. Cross-sectional regressions show that a one IQR increase in the average anomaly alignment of non-implementable anomalies is associated with a 0.08 percentage point decrease in the average annualized management fee. Panel regressions support this finding, indicating that a one IQR increase in current anomaly alignment of implementable anomalies is associated with a 0.05 percentage point decrease in annualized management fees in the subsequent month. This effect is economically meaningful, given that the average annualized management fee in the full sample is 0.44%.

Related Literature. We add to the literature discussing the decaying performance of return anomalies. [McLean and Pontiff \(2016\)](#) show that anomaly performance deteriorates after their publication in academic journals, and [Green et al. \(2017\)](#) document a general trend of decaying performance in return anomalies after 2003, with [Chen and Velikov \(2023\)](#) showing that anomalies fail to survive transaction costs in the form of effective spreads. [DeMiguel, Martin-Utrera, Nogales, and Uppal \(2020\)](#) show that the consideration of transaction costs changes the number of significant characteristics, and [Jensen et al. \(2025\)](#) estimate an “implementable” efficient frontier that directly depends on rebalancing costs. Using actual transactions of an institutional investor, [Frazzini et al. \(2018\)](#) show that effective transaction costs are well-explained by a quadratic price impact, which we use in this paper.

We also contribute to the literature that tries to understand the investment process of mutual funds. Closely related to our study, [Patton and Weller \(2020\)](#) estimate the all-

in trading costs of popular return anomalies from the exposure of mutual funds, and Johansson, Sabbatucci, and Tamoni (2025) replicate anomalies with ETFs and mutual funds and show that the resulting returns are smaller than on paper.

Finally, we add to the broad literature that applies machine learning methods to measure expected returns (Gu et al., 2020; Bali et al., 2023). Similar to Avramov, Cheng, and Metzker (2023) and Jensen et al. (2025), we argue in favor of incorporating information about investment frictions and transaction costs when evaluating the efficacy of these methods in asset pricing, and link these considerations to the de facto decision-making process of investment professionals.

2. Methodology

Our approach is best understood as a dependent double sort. For this, at each point in time t , we sort stocks in a first step into quintile portfolios based on one of the 153 characteristics c provided in the dataset of Jensen et al. (2023).² From this exercise we obtain the value weighted gross return of the high minus low portfolio sorted by characteristic c at time t and denote it by $r_{c,t+1}^{HL}$, with

$$r_{c,t+1}^{HL} = r_{c,t+1}^{High} - r_{c,t+1}^{Low} \quad (1)$$

where $r_{c,t+1}^{High}$ and $r_{c,t+1}^{Low}$ are the value weighted returns of the quintile portfolios with the highest and lowest values of characteristic c , respectively. While earlier studies found these anomaly characteristics to contain valuable investment advice in the sense that a hypothetical investor following the strategy outlined in Eq. (1) could have expected positive average risk adjusted returns, those strategies appear to have ceased to deliver in the more recent past. Green et al. (2017) show that most characteristics fail to uncover valuable information for future returns after 2003, an effect that is distinct from the post publication decay discussed in McLean and Pontiff (2016). Furthermore, Chen and Velikov (2023) show that most remaining characteristics with a significant gross return fail to deliver once transaction costs and fees are taken into account.

The second step in our methodology attempts to *revive* these anomalies by incorporating an estimate of expected net returns into the sorting procedure. Specifically, suppose

²For this exercise we adhere to the sorting direction from the original study as compiled by Jensen et al. (2023).

we have an estimate of the expected future net return for stock i , which we denote by $\mathbb{E}_t[r_{i,t+1}^{net}]$. Within each quintile portfolio formed on characteristic c , we sort stocks again according to $\mathbb{E}_t[r_{i,t+1}^{net}]$ into quintile portfolios. This procedure yields a set of 25 portfolios that are formed through dependent sorting. Our trading strategy focuses on the extreme portfolios implied by this double sorting procedure. On the long side, we select the portfolio that contains stocks with the highest values of $\mathbb{E}_t[r_{i,t+1}^{net}]$ within the quintile of stocks with the highest values of characteristic c . On the short side, we select the portfolio that contains stocks with the lowest expected net returns within the quintile of stocks with the lowest values of characteristic c . The return of the resulting strategy therefore reflects the performance difference between these two portfolios. Formally, let $r_{c,t+1}^{High,High}$ denote the value weighted return of the portfolio of stocks with both the highest values of characteristic c and the highest values of $\mathbb{E}_t[r_{i,t+1}^{net}]$. Analogously, $r_{c,t+1}^{Low,Low}$ denotes the value weighted return of the portfolio of stocks with both the lowest values of characteristic c and the lowest values of $\mathbb{E}_t[r_{i,t+1}^{net}]$. The return of our double sorted strategy, $r_{c,t+1}^{HL,Revived}$, is then given by

$$r_{c,t+1}^{HL,Revived} = r_{c,t+1}^{High,High} - r_{c,t+1}^{Low,Low}. \quad (2)$$

This construction isolates stocks that simultaneously exhibit extreme values along both sorting dimensions. The long leg consists of stocks with high values of characteristic c that are also assigned high expected net returns, whereas the short leg consists of stocks with low characteristic values and low expected net returns. By conditioning the anomaly portfolio on expected net returns, the strategy concentrates positions in stocks for which the anomaly signal and the expected net return estimate point in the same direction.

There are two key benefits of this approach. First, the dependent double sort ensures diversification of the final portfolios, as the number of stocks in each portfolio depends only on the size of the stock universe and the number of target portfolios. Second, irrespective of how the expected return signal, $\mathbb{E}_t[r_{i,t+1}^{net}]$, is obtained, the resulting portfolio maintains the interpretability of a characteristic sorted portfolio. This feature is particularly important for asset managers who are required to communicate their investment strategies to potential investors. While we will demonstrate how machine learning methods can be used to obtain estimates of expected future net returns, the portfolio construction framework mitigates some of the concerns related to the opaque nature of such models. Because the final portfolios remain organized along familiar characteristic dimensions, the investment strategy retains a transparent economic interpretation even when the un-

derlying expected return estimates are generated using complex predictive models. In the empirical analysis in Section 4, we show that mutual funds can, and likely do, exploit this mechanism to construct superior portfolios while maintaining the appearance of traditional characteristic based investment strategies.

Having established the double sorting framework, we next describe how expected gross returns, $\mathbb{E}_t[r_{i,t+1}]$, are estimated for each stock. In our setting, these estimates are obtained using machine learning methods and are part of the second sorting dimension of the strategy. Transaction costs (TC) are considered separately and enter additively, that is,

$$\mathbb{E}_t[r_{i,t+1}^{net}] = \mathbb{E}_t[r_{i,t+1}] - \mathbb{E}_t[TC]. \quad (3)$$

We describe how we estimate transaction costs in the paragraph thereafter.

Expected Returns. We know from a rapidly growing literature that tools from machine learning are suitable to recover expected returns of individual stocks (e.g. [Gu et al., 2020](#); [Beckmeyer and Wiedemann, 2025](#)) and other asset classes (e.g. [Bali et al., 2023](#)). Using these insights, we follow [Gu et al. \(2020\)](#) and express our estimate of future expected *gross* returns as

$$\mathbb{E}_t[r_{i,t+1}] = f(c_{i,t}). \quad (4)$$

where $f(\cdot)$ is a flexible function mapping today’s firm characteristics to expected future returns.³ As the authors show that neural networks outperform competing models in the task of stock return predictions, we follow their findings and use a feed-forward neural network with three hidden layers and the pyramidal layering scheme described in [Gu et al. \(2020\)](#) to estimate the mapping f . As a estimation of the neural network is computationally costly, we re-estimate the model on a yearly frequency. That is, we re-fit our machine learning estimator with information from up until the end of December each year and hold it fixed the entire next year. As our first out of sample period starts in January 2004, we use the preceding 23 years – starting in January 1980 – to estimate the first model. We use the first 70% of observations to train the model’s parameters and validate the model in the remaining 30% of months. We estimate 50 trials and use an equal weighted ensemble of the five best models for the final return prediction. For the predictions in 2005 we expand the training sample with the information from 2004 and

³As before, we use the dataset with information on 153 distinct firm-characteristics from [Jensen et al. \(2023\)](#) for this task.

re-estimate the model. We repeat the process until all years in our sample are exhausted. In a first step we only include this gross return estimate in our double-sorting exercise. As our transaction costs estimates are additive, we can separately study their impact on the overall performance.

Transaction Costs. Of course, a fund looking to invest by characteristic c will naturally also incorporate a notion of expected and realized transaction costs before allocating capital into a specific stock. For this component, we follow [Jensen et al. \(2025\)](#) and model the (expected and realized) dollar transaction costs of a trade with dollar-size vector τ_t as

$$TC_{t,i} = \frac{1}{2} \tau_t' \Lambda_{i,t} \tau_t, \quad (5)$$

where the factor of $\frac{1}{2}$ reflects the standard assumption that the realized cost is split equally between buyer and seller and $\Lambda_{i,t}$ is a diagonal matrix of price-impact coefficients, calibrated to the empirical estimates of [Frazzini et al. \(2018\)](#). Specifically, [Frazzini et al. \(2018\)](#) document that trading 1% of a stock's daily dollar volume moves its price by approximately 0.1%, implying a linear price-impact coefficient of $0.1\%/1\% = 0.1$ per unit of daily dollar volume. Consequently,

$$\Lambda_{i,t} = \frac{0.2}{V_{i,t}}, \quad (6)$$

where $V_{i,t}$ is the expected daily dollar volume of stock i on day t .⁴ Intuitively, stocks with higher turnover are cheaper to trade: a larger $V_{i,t}$ implies a smaller price-impact coefficient and therefore lower transaction costs for a given trade size. At the same time, the quadratic form in (5) implies that costs are convex in trade size, capturing the empirical regularity that large orders face increasing marginal price impact and therefore penalizing excessive concentration in illiquid stocks. Following [Frazzini et al. \(2018\)](#) and [Jensen et al. \(2025\)](#), we proxy for expected daily volume using the average daily dollar volume over the trailing six months.

⁴To see how (6) arises, note that a trade of size τ moves the price by $0.001 \times (\tau/0.01V_{i,t}) = 0.1 \tau/V_{i,t}$, so the dollar cost of the trade is $\tau \times 0.1 \tau/V_{i,t} = 0.1 \tau^2/V_{i,t}$. Matching this to $\frac{1}{2} \Lambda_{i,t} \tau^2$ immediately gives $\Lambda_{i,t} = 0.2/V_{i,t}$.

3. Reviving Anomalies

Ex Post Analysis. In the first step, we demonstrate the effectiveness of our machine learning augmentation of standard factor portfolios. For this reason, we abstract from expected transaction costs and focus on a double sort based on the respective factor and expected returns. We first show that augmenting characteristic-sorted high-minus-low portfolios with the expected returns of portfolio constituents severely improves their out-of-sample performance over the period from January 2004 to December 2023. In Figure 1, the top panel illustrates the average improvement in annualized gross returns, while the bottom panel reports the corresponding average t -statistic from a comparison of the standard *hml* return with the expected return-augmented *hml* return. We cluster characteristics by the themes identified by Jensen et al. (2023) and report average improvements within each theme.

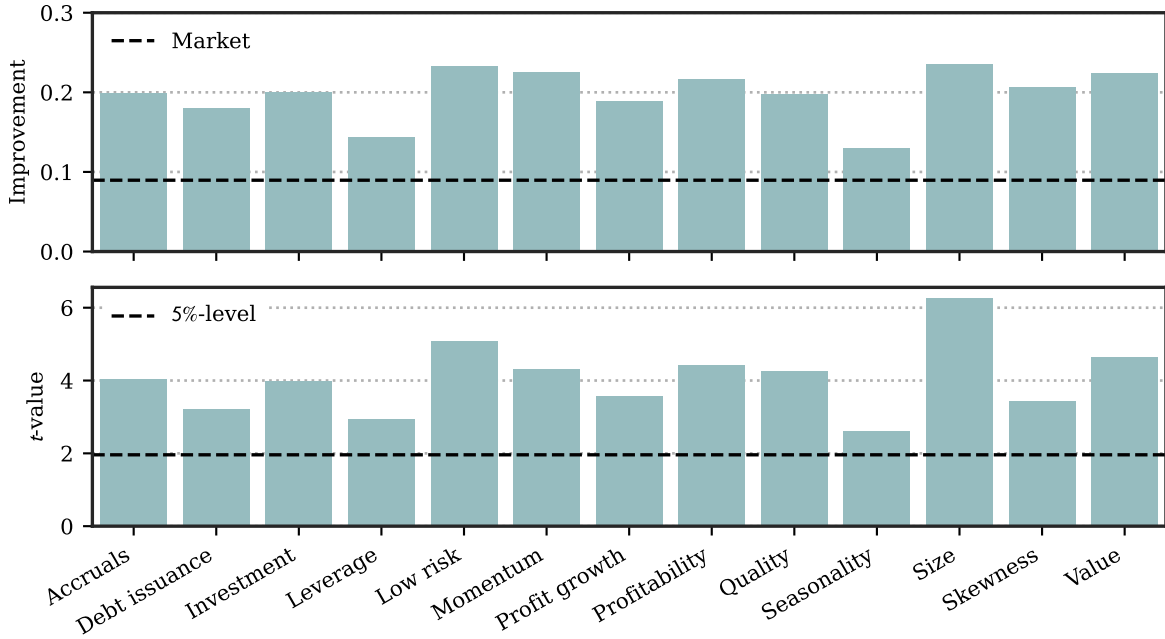


Fig. 1. Average Improvement per Anomaly Theme

The figure shows the mean gross return improvement (top) and the mean t -value of this improvement (bottom) across anomalies of different themes. We measure the improvements as the difference between the value-weighted return of the $\mathbb{E}[r]$ -adjusted and the unadjusted factor portfolios. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). The 13 anomaly themes are identified by Jensen et al. (2023) and cover all 153 characteristic-based anomalies. The testing sample spans the years 2004 through 2023.

We observe substantial economic and statistically significant improvements across all

characteristic themes, ranging from +12.5% for characteristics capturing the seasonality of stock returns to +22% for size and low-risk characteristics. Notably, the average improvement within each of the 13 themes generates gross returns exceeding those of a market-wide investment over the same period, highlighting the benefits of incorporating stock-specific expected returns into systematic investment strategies. The t-statistics reported in the lower panel are consistently large, ranging from 2.3 for seasonality to over 6 for size. Many of these t-values exceed the elevated threshold of 3 recommended by [Harvey, Liu, and Zhu \(2015\)](#) to mitigate concerns of multiple hypothesis testing and p-hacking.

While the gross-of-fees improvements depicted in Figure 1 are great, they do not indicate whether an investor could have profited from such an investment, as they ignore transaction costs. Price impact costs for large institutions are known to increase with invested capital, which we acknowledge in the definition of transaction costs in Eq. (6). Institutional investors hold approximately half of the U.S. equity market ([Gârleanu and Pedersen, 2022](#)), and [Edelen, Evans, and Kadlec \(2007\)](#) show that two thirds of transaction costs for mutual funds are due to price impact costs, far outweighing proportional costs such as bid-ask spreads. We therefore examine how many of the 153 characteristic-based anomaly portfolios would have generated significant out-of-sample *net* returns for institutions with different AUM. We assume four terminal fund sizes in December 2023. These four levels are inspired by the distribution of fund sizes in the CRSP Mutual Fund Database, roughly corresponding to the 20th, 40th, 60th, and 80th percentiles. We consider only mutual funds with an equity share of at least 50% (based on holdings matched to our JKP sample) and a minimum of 10 matched stocks. The distribution of fund sizes is reported in Appendix A. Following [Jensen et al. \(2025\)](#), we discount fund sizes back in time using the market return, which approximately reflects the general growth of the industry.

Table 1 reports the results. The first panel (Unadjusted) presents the number of anomaly portfolios that generate significant net returns between 2004 and 2023 when stocks are sorted into quintiles by characteristic c . These portfolios are constructed without incorporating either expected returns or expected transaction costs. As expected, the number of significant anomalies decreases with the fund size. A micro fund managing 40mn USD has 17 characteristic-sorted anomaly portfolios at its disposal, a small (medium-sized) fund managing 200mn (500mn) USD has 10 (3), and a large fund managing 1.6bn USD has only a single anomaly available. The average portfolio returns are broadly compa-

rable, ranging from 6 – 8% per year, with Sharpe ratios around 0.6, roughly matching the market over the same period. Maximum drawdowns are moderate, amounting to approximately one-third of the invested capital.

Table 1: Average Anomaly Performance

The table shows the average net performance of anomalies that are *ex post* significantly positive at the 5% level, accounting for price impact costs for different fund sizes. The results show the number of significantly positive anomalies (N), the mean of net returns (Net ret), the standard deviation of net returns (Std), the Sharpe ratio (Sharpe), and the maximum drawdown (Max. DD). All 153 characteristic-based anomalies documented in [Jensen et al. \(2023\)](#) are considered. We calculate the value-weighted net performance for (i) unadjusted, (ii) $\mathbb{E}[r]$ -adjusted, and (iii) $\mathbb{E}[r]$ - & TC-adjusted factor portfolios. Expected returns are measured out-of-sample, following the neural network approach in [Gu et al. \(2020\)](#). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following [Jensen et al. \(2025\)](#) and [Frazzini et al. \(2018\)](#). The four fund sizes (40, 200, 500, and 1600), measured in million USD, are motivated by the distribution of fund sizes in the CRSP Mutual Fund Database and approximately correspond to the 20th, 40th, 60th, and 80th percentiles. The fund sizes are measured in 2023 and discounted back in time with the market return. The testing sample spans the years 2004 through 2023.

	Unadjusted				$\mathbb{E}[r]$ -adjusted				$\mathbb{E}[r]$ - & TC-adjusted			
	40	200	500	1600	40	200	500	1600	40	200	500	1600
N	17	10	3	1	126	73	9	0	126	101	81	41
Net ret	0.08	0.08	0.07	0.06	0.21	0.18	0.16		0.18	0.14	0.11	0.09
Std	0.13	0.14	0.11	0.11	0.26	0.26	0.28		0.24	0.20	0.17	0.14
Sharpe	0.57	0.60	0.62	0.59	0.82	0.70	0.57		0.75	0.71	0.68	0.62
Max. DD	-0.32	-0.34	-0.26	-0.31	-0.55	-0.58	-0.60		-0.51	-0.44	-0.39	-0.34

Adjusting factor exposures by the expected gross return component of the constituent stocks greatly amplifies the number of anomaly portfolios exhibiting *ex post* significant net returns for micro- to medium-sized funds. For micro funds, 126 of the 153 anomaly portfolios display significant net returns; for small funds, 73 portfolios do so; and for medium-sized funds, 9 portfolios. Only for the largest mutual funds in the CRSP Mutual Fund Database does a mere expected return adjustment fail to uncover additional investment opportunities. Indeed, not a single anomaly portfolio exhibits significant returns after incorporating price impact costs for a large fund. Average net returns decline slightly with fund size, ranging from 21% per year for micro funds to 16% for medium-sized funds. Realized net Sharpe ratios follow a similar pattern, decreasing from 0.82 to 0.57, as return standard deviations are broadly comparable. Notably, maximum drawdowns increase substantially after adjusting for the expected return component of the stocks, ranging between -55% and -60%, highlighting that investing in high-expected return stocks intensifies the risk of “being wrong”.

Once we augment the construction of anomaly portfolios by incorporating both the expected gross returns of the constituent stocks and their expected transaction costs, we observe a substantial increase in the number of viable investment portfolios for medium- to large-sized funds. While the number of significant anomalies is roughly comparable for micro and small funds, it rises sharply for medium-sized and large funds. Considering only expected returns identifies 9 (0) anomaly portfolios with significant net returns for a medium-sized (large) fund. Including expected transaction costs increases these numbers to 81 (41) anomaly portfolios. Average net returns are 18% per year for a micro fund, decreasing to 9% per year for a large fund. The standard deviation of returns declines with fund size, resulting in realized net Sharpe ratios that are broadly comparable across micro to medium-sized funds, at 0.75 and 0.68, respectively. Interestingly, maximum drawdowns are also lower than when augmenting only by expected returns, suggesting that several investment opportunities that appear promising on paper are not investable for medium-sized to large funds.

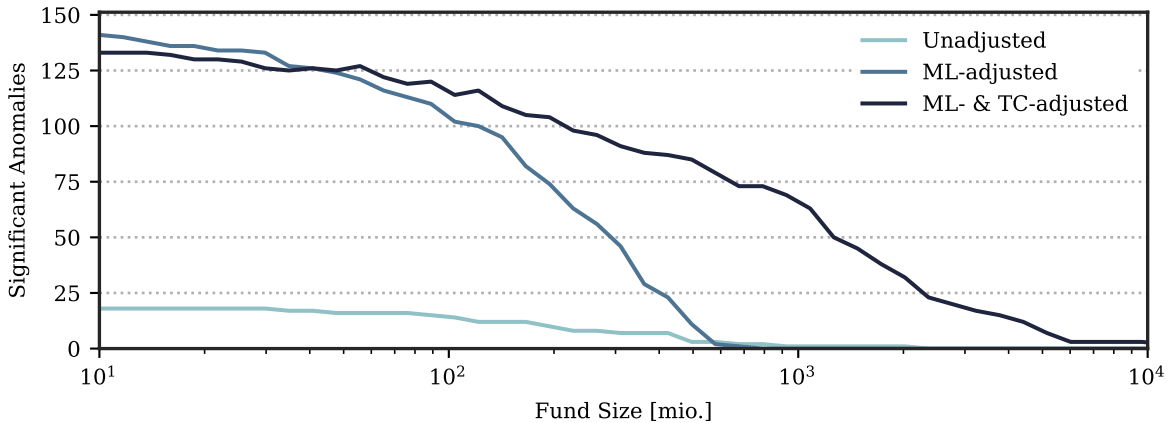


Fig. 2. Number of Significant Anomalies per Fund Size

The figure shows the number of anomalies that have ex post significantly positive net returns, accounting for price impact cost for different fund sizes. All 153 characteristic-based anomalies documented in [Jensen et al. \(2023\)](#) are considered. We calculate the value-weighted net returns for (i) unadjusted, (ii) $\mathbb{E}[r]$ -adjusted, and (iii) $\mathbb{E}[r]$ - & TC-adjusted factor portfolios. Expected returns are measured out-of-sample, following the neural network approach in [Gu et al. \(2020\)](#). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following [Jensen et al. \(2025\)](#) and [Frazzini et al. \(2018\)](#). The interval of fund sizes $[10^1, 10^4]$, measured in million USD, covers approximately 95% of all fund sizes in the CRSP Mutual Fund Database in 2023. The fund sizes are discounted back in time with the market return. The testing sample spans the years 2004 through 2023.

In Figure 2, we report the number of anomalies with significant out-of-sample net returns across three construction methods: unadjusted, $\mathbb{E}[r]$ -adjusted, and $\mathbb{E}[r]$ - and TC-adjusted, for funds of varying sizes. The figure highlights the benefits of augmenting

portfolio construction with the expected gross returns of the constituent stocks. For medium-to-large funds, further adjustment for expected price impact costs becomes crucial. Once a fund manages more than approximately 200mn USD, incorporating expected transaction costs is as important as considering expected gross returns. Together, these adjustments reveal systematic investment opportunities even for the largest funds in the CRSP Mutual Fund Database.

Table 2 shows the number of anomaly portfolios exhibiting significant net returns, categorized by the 13 characteristic themes proposed by Jensen et al. (2023). Without any adjustments, significant net returns are heavily clustered: 10 of the 17 quality-based portfolios remain significant after an out-of-sample test that accounts for price impact costs for a micro fund. Similarly, 2 of the 5 accruals-based portfolios survive. For larger funds, these figures naturally decline: managing 500mn USD, only 2 quality-based portfolios remain significant, while none of the accruals-based portfolios do.

Both augmentations substantially improve the landscape and diversity of investable anomaly portfolios. A small fund can now access 9 of 13 profit growth-based portfolios, all 8 momentum portfolios, 15 of 17 quality-based portfolios, 14 of 22 investment-based portfolios, and all 12 profitability-based anomalies. Most of the remaining characteristic themes are similarly well represented after these adjustments. The resulting diversification benefits for the combined investable portfolio are potentially large, as different sources of investment information are incorporated. For the largest funds in our sample, managing 1.6bn USD as of December 2023, a total of 41 anomalies generate significant net returns. Quality-based anomalies exhibit the strongest performance, followed by accrual-, debt issuance-, and skewness-based portfolios. Notably, none of the size- or value-based anomalies are revived for these larger funds, and only a single anomaly in each of the profit growth-, momentum-, and leverage-based themes survives, highlighting their dependence on illiquid stocks with small market capitalizations.

Ex Ante Analysis. While the previous analyses demonstrate that the augmented portfolio construction techniques substantially increase the number of investable anomaly portfolios, they assess a portfolio’s investability only ex post. This exercise is important because it shows that even large mutual funds could have profited meaningfully from systematic investment opportunities documented in the academic literature, particularly when leveraging recent advances in the predictability of individual stock returns. Our framework also allows us to evaluate how many of the 153 anomaly portfolios are *expected*

Table 2: Number of Significant Anomalies per Anomaly Theme and Fund Size

The table shows the number of anomalies that have ex post significantly positive net returns at the 5% level, accounting for price impact costs for different fund sizes. The 13 themes are identified by Jensen et al. (2023) and cover all 153 characteristic-based anomalies. We calculate the value-weighted net return for (i) unadjusted, (ii) $\mathbb{E}[r]$ -adjusted, and (iii) $\mathbb{E}[r]$ - & TC-adjusted factor portfolios. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following Jensen et al. (2025) and Frazzini et al. (2018). The four fund sizes (40, 200, 500, and 1600), measured in million USD, are motivated by the distribution of fund sizes in the CRSP Mutual Fund Database and approximately correspond to the 20th, 40th, 60th, and 80th percentiles. The fund sizes are measured in 2023 and discounted back in time with the market return. The testing sample spans the years 2004 through 2023.

Theme	N	Unadjusted				$\mathbb{E}[r]$ -adjusted				$\mathbb{E}[r]$ - & TC-adjusted			
		40	200	500	1600	40	200	500	1600	40	200	500	1600
Profit growth	13	1	1	1	0	12	3	0	0	12	9	5	1
Momentum	8	0	0	0	0	8	8	1	0	8	8	4	1
Quality	17	10	8	2	1	16	16	5	0	15	15	16	13
Value	13	1	0	0	0	13	6	0	0	12	5	1	0
Investment	22	0	0	0	0	19	10	0	0	19	14	12	4
Low risk	21	0	0	0	0	15	11	1	0	17	12	8	2
Profitability	12	1	0	0	0	12	8	1	0	12	12	9	5
Leverage	11	0	0	0	0	5	1	0	0	5	3	3	1
Debt issuance	6	1	1	0	0	6	2	1	0	6	5	5	3
Accruals	5	2	0	0	0	5	4	0	0	5	4	4	4
Seasonality	14	0	0	0	0	8	4	0	0	8	8	9	4
Size	5	0	0	0	0	1	0	0	0	1	0	0	0
Skewness	6	1	0	0	0	6	0	0	0	6	6	5	3
All	153	17	10	3	1	126	73	9	0	126	101	81	41

to generate a positive net return by comparing the included stocks' expected returns with their anticipated transaction costs, which we define as *implementable* anomalies. This ex ante analysis thus closely mirrors the decision-making process of a fund manager. Figure 3 depicts the number of anomalies per month expected to deliver a positive net return over the subsequent month for the four fund sizes considered.

For micro to medium-sized funds, we find that more than 140 anomalies are expected to generate a positive net return over the following month, with only minor fluctuations over time. This suggests that the machine learning estimator anticipates outperforming expected transaction costs for many anomalies. For the largest mutual funds in CRSP, the number of anomalies with a positive expected net return fluctuates between 120 and 140 until 2016 and declines thereafter, with a further drop observed in 2023. It is important to note that this exercise does not assess the statistical significance of these positive

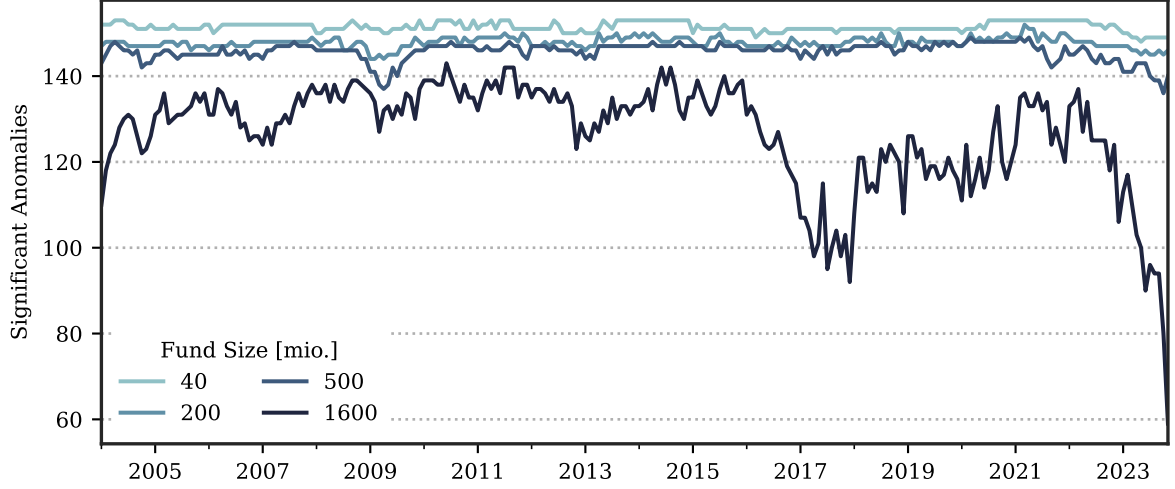


Fig. 3. Ex Ante Analysis of Implementable Anomalies per Fund Size

The figure shows the number of anomalies that are ex ante expected to yield positive net returns after our expected return and transaction cost augmentations. All 153 characteristic-based anomalies documented in Jensen et al. (2023) are considered. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following Jensen et al. (2025) and Frazzini et al. (2018). The four fund sizes (40, 200, 500, and 1600), measured in million USD, are motivated by the distribution of fund sizes in the CRSP Mutual Fund Database and approximately correspond to the 20th, 40th, 60th, and 80th percentiles. The fund sizes are measured in 2023 and discounted back in time with the market return. The testing sample spans the years 2004 through 2023.

expected net returns. Nevertheless, comparing these results with the ex post analysis in Table 1 reveals a strong alignment between our ex ante measure of investability and the realized ex post profitability.

Table 3 reports performance metrics over a 20-year out-of-sample period for an equally weighted investment in anomalies expected to generate a positive net return, across funds of different AUM levels. The literature documents that outperforming a simple 1/N strategy is challenging (DeMiguel, Garlappi, and Uppal, 2009). We therefore use this as a baseline and leave potential benefits from optimizing across the achievable set of anomalies to future research.

Starting with a micro fund, the average net return of this 1/N strategy is 16.0% per year, with a Sharpe ratio of 1.01 and a maximum drawdown of -34.4%. As fund size increases, overall profitability declines. A small (medium-sized) fund has averaged net returns of 12.0% (9.2%) after fees, with Sharpe ratios of 0.90 (0.85). For a large fund, the average net return falls to 5.4%, with a Sharpe ratio of 0.64. Notably, across all fund sizes, the strategy exhibits negative exposure to market returns ($\beta < 0$), generating substantial

Table 3: OOS Analysis of 1/N Investment Strategy

The table shows the mean net return (Net ret), the standard deviation of net returns (Std), the Sharpe ratio (Sharpe), the maximum drawdown (Max. DD), and the CAPM parameters (α, β) for a 20-year out-of-sample period of a 1/N investment in anomalies that are ex ante expected to generate a positive net return for different fund sizes. All 153 characteristic-based anomalies documented in Jensen et al. (2023) are considered. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following Jensen et al. (2025) and Frazzini et al. (2018). The four fund sizes (40, 200, 500, and 1600), measured in million USD, are motivated by the distribution of fund sizes in the CRSP Mutual Fund Database and approximately correspond to the 20th, 40th, 60th, and 80th percentiles. The fund sizes are measured in 2023 and discounted back in time with the market return. The testing sample spans the years 2004 through 2023.

Fund size [mio.]	Net ret	Std	Sharpe	Max. DD	CAPM	
					α	β
40	0.160	0.158	1.010	−0.344	0.203***	−0.480
200	0.120	0.133	0.904	−0.303	0.155***	−0.388
500	0.092	0.108	0.851	−0.278	0.120***	−0.314
1600	0.054	0.085	0.639	−0.247	0.077***	−0.255

and highly significant alphas. These findings imply substantial diversification benefits of augmenting a long-only investment with the set of implementable anomalies.

4. Mutual Funds

We have proposed two simple augmentations that vastly increase the number of implementable anomaly portfolios for funds of all sizes. In this section, we examine whether funds incorporate information regarding which anomalies are ex ante implementable into their investment decisions. Specifically, we demonstrate that funds strategically tilt their portfolios toward anomalies that are more likely to be implementable net of price impact costs, conditional on their AUM. We formalize this behavior through a new measure, *anomaly alignment*.

Anomaly Alignment. We merge the CRSP Mutual Fund Database with the 153 firm-level characteristics documented in Jensen et al. (2023), which are constructed using information from CRSP and Compustat. To each applicable holding, we add information about the 153 firm-level characteristics. Every month, we calculate the funds’ AUM and their respective equity share. Since we are investigating the profitability of equity

anomaly portfolios, we restrict the sample to funds with an equity share of at least 50% based on holdings matched to our JKP dataset, and with at least 10 matched stocks.

We define the anomaly alignment of fund f with respect to each anomaly a at time t as follows. For each anomaly and point in time, we construct a target portfolio that jointly incorporates expected return forecasts from our machine learning model and expected transaction costs estimated via the quadratic price impact model. Following prior methodology, the target portfolio is obtained through a double sort on the respective anomaly and expected returns net of transaction costs. Because transaction costs are a function of fund size, we compute target portfolios over a grid of 100 fund sizes, linearly spaced according to the empirical distribution of fund sizes in the sample (see Figure A). For computational efficiency, each fund is then assigned the target portfolio corresponding to the nearest grid size. Within each target portfolio, each stock i is assigned a dummy variable $D_{i,a,t}$ which equals 1 if the stock is included in the long leg, -1 if in the short leg, and 0 otherwise. The fund's anomaly alignment with respect to anomaly a is then computed as the sum of the portfolio weights $w_{f,i,t}$ multiplied by the corresponding dummy values, based on the fund's actual holdings at time t :

$$\text{ANOMALY ALIGNMENT}_{f,a,t} = \sum_i w_{f,i,t} \cdot D_{i,a,t}. \quad (7)$$

Economically, a higher anomaly alignment indicates that a fund more closely tracks our augmented anomaly portfolio.

Proof of Principle. Using the previous double sort, we estimate which of the 153 anomaly portfolios are likely implementable net of fees, taking into account both expected returns and anticipated transaction costs. At each point in time, each fund has 153 different anomaly alignments, which are classified as either implementable or non-implementable. In a final step, we compute the mean anomaly alignment across anomalies, distinguishing between those identified as implementable and those classified as non-implementable.

To examine whether mutual funds actively tilt their portfolios toward implementable anomalies, we estimate the following panel regression:

$$\text{ANOMALY ALIGNMENT}_{f,t} = \mathbb{1}_{f,t} + \text{TE} + \text{FE}, \quad (8)$$

where $\mathbb{1}_{f,t}$ is an indicator variable equal to 1 when the anomaly alignment corresponds to the mean of implementable anomalies, and 0 when it corresponds to the mean of non-implementable anomalies. TE and FE denote time- and fund-fixed effects, respectively. Standard errors are clustered by time and entity.

The results in Table 4 indicate that funds align their holdings towards implementable anomalies and away from anomalies unachievable to them. Because the selection of investable anomalies depends on fund size, this approach implicitly accounts for differences in AUM. On average, the anomaly alignment for implementable anomalies is 0.05 higher than for non-implementable anomalies without fixed effects. Incorporating time-fixed effects increases the difference to 0.11, while including both time- and fund-fixed effects further raises it to 0.12. Overall, the findings provide clear evidence that mutual funds systematically align their portfolios with implementable anomalies.

Table 4: Panel Regression of Anomaly Alignment

The table shows panel regressions of anomaly alignment on an indicator function, accounting for time- and fund-fixed effects. In this setting, anomaly alignment refers to the mean anomaly alignment of either implementable or non-implementable anomalies. The indicator function equals 1 if the anomalies are implementable and 0 if the anomalies are non-implementable for a given fund and time. Standard errors are clustered by time and entity. The testing sample spans the years 2004 through 2023.

	Anomaly Alignment		
	(1)	(2)	(3)
$\mathbb{1}$	0.053	0.109	0.120
t -value	***	***	***
TE		x	x
FE			x

Regression Setting. To examine whether fund characteristics are influenced by anomaly alignment, we use regression analyses. Consistent with the previous analysis, we distinguish between the mean anomaly alignment of implementable and non-implementable anomalies. This allows us to assess whether changes in the anomaly alignment of implementable anomalies have a different effect on fund characteristics compared with changes in the anomaly alignment of non-implementable anomalies. We begin with cross-sectional regressions in which the mean of a fund characteristic is regressed on the fund’s mean anomaly alignment:

$$\text{FUND CHARACTERISTIC}_f = \text{Intercept} + \text{Anomaly Alignment}_f + \text{Controls}, \quad (9)$$

where $\text{FUND CHARACTERISTIC}_f$ represents an average attribute of fund f (e.g., fund returns or management fees) over time, while $\text{Anomaly Alignment}_f$ denotes the fund's average anomaly alignment across anomalies and time. We estimate two separate regressions: one for implementable anomalies and one for non-implementable anomalies. Standard errors are heteroskedasticity-robust (HC3). These cross-sectional regressions provide insights into how average fund characteristics relate to the average anomaly alignment.

Next, we extend this analysis using panel regressions to capture the time-varying nature of anomaly alignment:

$$\text{FUND CHARACTERISTIC}_{f,t} = \text{Anomaly Alignment}_{f,t-1} + \text{TE} + \text{FE} + \text{Controls}, \quad (10)$$

where $\text{FUND CHARACTERISTIC}_{f,t}$ denotes a specific attribute of fund f at time t , and $\text{Anomaly Alignment}_{f,t}$ represents the average anomaly alignment across anomalies for fund f at time $t - 1$. Analogously, we estimate two separate regressions, one for implementable anomalies and one for non-implementable anomalies. Standard errors are clustered at the fund and time levels. This panel regression framework enables us to assess how anomaly alignment predicts future fund characteristics.

Theoretically, our measure is unbounded. Therefore, we rely on its empirical distribution to provide an economic interpretation of the regression coefficients. In the full sample, the mean anomaly alignment across both implementable and non-implementable anomalies is 0.01, with a standard deviation of 0.10. The minimum and maximum values of the mean anomaly alignment are -0.36 and 0.57, respectively, and the 25th, 50th, and 75th percentiles are -0.05, 0.02, and 0.07. To evaluate the economic significance of the regression coefficients, we use its IQR of 0.11. Summary statistics for mutual fund characteristics are reported in Appendix B.

Fund Returns. As a first fund characteristic, we examine monthly returns to assess the impact of anomaly alignment on fund performance. Specifically, we investigate how monthly returns are affected when fund managers indirectly follow our approach by implementing anomaly portfolios that account for expected returns and transaction costs. In Table 5, we regress the mean fund return on the mean anomaly alignment for both implementable and non-implementable anomalies. For implementable anomalies, the results indicate that anomaly alignment has a significantly positive effect on fund returns.

Without controls, a one IQR increase in anomaly alignment raises the average monthly return by 0.62 percentage points, increasing to 0.71 percentage points after including controls. This effect is economically meaningful as it translates into a difference of 7.39 and 8.48 percentage points in annualized returns, respectively.⁵ Thus, funds that indirectly invest in implementable anomalies by accounting for expected returns and transaction costs achieve significantly higher returns on average. In contrast, investments in non-implementable anomalies do not produce a significant effect on average returns. These findings highlight the importance of both expected returns and transaction costs, as positions in non-implementable anomalies do not translate into higher realized returns.

Table 5: Cross-Sectional Regression of Fund Returns

The table shows cross-sectional regressions of the mean fund return on its mean anomaly alignment for both implementable and non-implementable anomalies. Controls include the log of the fund’s total net assets (`log_mtna`), its equity share based on the percentage of the holdings matched with our JKP sample (`percent_matched`), the management fees (`mgmt_fee`), the expense ratio (`exp_ratio`) and the turnover ratio (`turn_ratio`). Standard errors are heteroskedasticity-robust (HC3). The testing sample spans the years 2004 through 2023.

	Implementable		Not Implementable	
	(1)	(2)	(1)	(2)
Intercept	0.006 ***	0.006 **	0.008	−0.001
An. Align.	0.054 ***	0.062 ***	0.001	−0.011
Controls	x		x	

In Table 6, we extend the cross-sectional analysis using predictive panel regressions. Specifically, we examine whether funds with higher reported anomaly alignment achieve higher subsequent returns. To this end, we regress each fund’s return on its lagged anomaly alignment for both implementable and non-implementable anomalies. Consistent with the cross-sectional results, the effects differ between implementable and non-implementable anomalies. For implementable anomalies, a one IQR increase in anomaly alignment raises subsequent monthly returns by 0.42 percentage points, increasing to 0.50 percentage points after including controls. This effect is economically meaningful as it translates into a difference of 5.06 and 6.02 percentage points in annualized returns, respectively. In contrast, the effects for non-implementable anomalies are insignificant,

⁵Effect sizes are calculated using the IQR of anomaly alignment, which equals 0.114 for the full sample (Appendix B). The corresponding monthly return effect is $0.054 \times 0.114 = 0.62\%$, and the annualized effect is $0.054 \times 0.114 \times 12 = 7.39\%$.

underscoring the role of expected returns and transaction costs highlighted in the cross-sectional analysis.

Table 6: Panel Regression of Fund Returns

The table shows panel regressions of the fund’s return on its lagged anomaly alignment for both implementable and non-implementable anomalies. Controls include the log of the fund’s total net assets (`log_mtna`), its equity share based on the percentage of the holdings matched with our JKP sample (`percent_matched`), the management fees (`mgmt_fee`), the expense ratio (`exp_ratio`), and the turnover ratio (`turn_ratio`). Standard errors are clustered by time and entity. All specifications include time-fixed effects. The testing sample spans the years 2004 through 2023.

	Implementable		Not Implementable	
	(1)	(2)	(1)	(2)
An. Align.	0.037 ***	0.044 ***	0.006	0.014
Controls		x		x

Management Fees. As a second fund characteristic, we examine management fees to assess the influence of anomaly alignment on fund managers’ compensation. In Table 7, we regress each fund’s mean management fee on its mean anomaly alignment, distinguishing between implementable and non-implementable anomalies. For implementable anomalies, anomaly alignment has no significant effect on management fees. In contrast, for non-implementable anomalies, a higher anomaly alignment is associated with lower management fees on average. Without controls, a one IQR increase in anomaly alignment reduces the average management fee by 0.12 percentage points, and the effect remains economically meaningful at 0.08 percentage points after including controls, given that the mean management fee in the full sample is 0.44%.⁶ These results indicate that funds indirectly investing in non-implementable anomalies tend to charge lower management fees. The findings suggest that investors are aware of which anomalies are implementable for a given fund, and that funds failing to consider this distinction empirically face constraints on their ability to charge higher fees.

In Table 8, we extend the cross-sectional analysis using predictive panel regressions to examine whether funds with higher anomaly alignment subsequently charge different management fees. Specifically, we regress each fund’s management fee on its lagged anomaly alignment for both implementable and non-implementable anomalies. Consistent with

⁶Effect sizes are calculated using the IQR of anomaly alignment, which equals 0.114 for the full sample (Appendix B). The corresponding management fee effect is $-1.033\% \times 0.114 = -0.12\%$.

Table 7: Cross-Sectional Regression of Management Fees

The table shows regressions of the mean fund management fee on its mean anomaly alignment for both implementable and non-implementable anomalies. Controls include the log of the fund's total net assets (`log_mtna`), its equity share based on the percentage of the holdings matched with our JKP sample (`percent_matched`), the fund's return (`mret`), and the turnover ratio (`turn_ratio`). Standard errors are heteroskedasticity-robust (HC3). The testing sample spans the years 2004 through 2023.

	Implementable		Not Implementable	
	(1)	(2)	(1)	(2)
Intercept	0.380 ***	0.538 ***	0.328	0.722 ***
An. Align.	-0.153	-0.218	-1.033 ***	-0.704 ***
Controls		x		x

the cross-sectional results, the effects differ across the two groups. For implementable anomalies, anomaly alignment has no significant effect on management fees. In contrast, for non-implementable anomalies, a one IQR increase in anomaly alignment reduces subsequent management fees by 0.10 percentage points, with the effect remaining at 0.05 percentage points after including controls. Overall, these results indicate that higher anomaly alignment predicts lower subsequent management fees for non-implementable anomalies, consistent with investors being aware of investment constraints at the fund level.

Table 8: Panel Regression of Management Fees

The table shows panel regressions of the fund's management fee on its lagged anomaly alignment for both implementable and non-implementable anomalies. Controls include the log of the fund's total net assets (`log_mtna`), its equity share based on the percentage of the holdings matched with our JKP sample (`percent_matched`), the fund's return (`mret`), and the turnover ratio (`turn_ratio`). Standard errors are clustered by time and entity. All specifications include time-fixed effects. The testing sample spans the years 2004 through 2023.

	Implementable		Not Implementable	
	(1)	(2)	(1)	(2)
An. Align.	-0.150	-0.097	-0.861 ***	-0.401 ***
Controls		x		x

5. Robustness

To ensure that the return improvements across the 153 anomalies are not driven by investing in the same set of stocks after augmenting by the constituents’ expected returns, Figure 4 compares the commonality in the anomalies’ returns before and after incorporating expected returns into the portfolio construction.

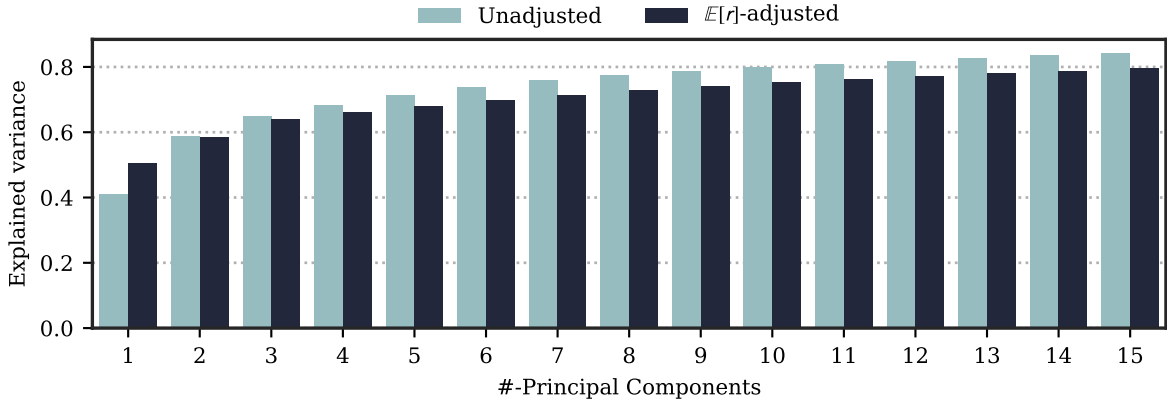


Fig. 4. Principal Component Analysis of Anomaly Returns

The figure shows the cumulative explained variance of the value-weighted gross return of unadjusted and $\mathbb{E}[r]$ -adjusted factor portfolios for a varying number of principal components. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). All 153 characteristic-based anomalies from Jensen et al. (2023) are considered. The testing sample spans the years 2004 through 2023.

To this end, we estimate the first 15 principal components of the out-of-sample returns for both the unadjusted and the expected return-adjusted anomaly portfolios, and compute the fraction of total variance explained by the first p principal components. While the first principal component accounts for a larger share of variation after the adjustment (increasing from approximately 40% to 50%), the cumulative variance explained by the first two components remains virtually unchanged. As the number of components increases, however, the augmented return series are *less* well captured by the first p principal components. This pattern suggests that the adjusted anomalies span a more heterogeneous set of investment opportunities.

Another potential concern is that our approach effectively resembles a dependent double sort into 25 portfolios. This procedure obviously differs from a single sort, raising the possibility that comparisons to a standard quintile sort may be inappropriate. In Figure 5, we address this concern by showing that our approach outperforms conventional

characteristic-sorted portfolios even when restricting attention to the extreme values of the target characteristic. Even highly concentrated portfolios continue to underperform our approach by a substantial margin.

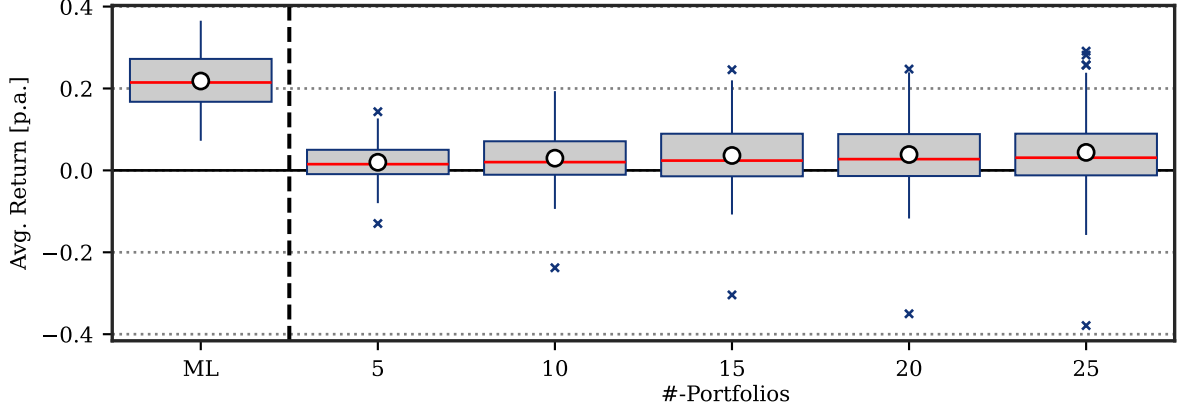


Fig. 5. Effect of Portfolio Concentration

The figure shows the distribution of mean value-weighted gross returns for the 153 different characteristic-sorted long-short portfolios (Jensen et al., 2023). It compares E[r]- & TC-adjusted factor portfolios (ML) to a single characteristic-sorted portfolio with various concentration. We show results for sorts into 5, 10, 15, 20 and 25 portfolios. Expected returns are measured out-of-sample, following the neural network approach in Gu et al. (2020). Expected transaction costs are determined for each fund size by a scale-dependent price impact model, following Jensen et al. (2025) and Frazzini et al. (2018). The testing sample spans the years 2004 through 2023.

6. Conclusion

In this paper, we introduce a novel framework to construct anomaly portfolios that jointly incorporates machine learning return forecasts and scale-dependent price impacts. Our approach addresses the prevailing view that most anomalies become insignificant once implementation costs are taken into account by showing that a large set of anomalies can be revived across a wide range of fund sizes. We find that expected returns are more important for anomaly portfolios of smaller funds, whereas expected transaction costs are especially relevant for larger funds. Across anomaly themes, our framework performs particularly well for quality- and accrual-based strategies. In contrast, size-based anomalies cannot be revived for most fund sizes, consistent with their reliance on illiquid, small-cap stocks. Importantly, because our method operates as an overlay on standard factor portfolios, it preserves their interpretability while selecting only the most promising investment opportunities within each factor portfolio. Investors seeking

to exploit return anomalies can therefore readily incorporate our framework into existing strategies.

Building on this framework, we introduce the concept of implementable anomalies, defined as anomalies that are expected to generate positive net returns conditional on fund size and time. A simple 1/N investment strategy in implementable anomalies delivers economically meaningful out-of-sample Sharpe ratios ranging from 0.64 to 1.01 across fund sizes. The resulting portfolios exhibit negative market exposure ($\beta < 0$) and generate large and significant alphas, implying substantial diversification benefits relative to a long-only market investment. We leave the question of optimal portfolio allocation between implementable anomalies and a long-only market investment to future research.

Finally, we provide evidence that mutual funds behave in a manner consistent with our framework, as they tilt their holdings toward implementable anomalies and away from non-implementable ones. Mutual funds with higher exposure to implementable anomalies earn significantly higher net returns, in line with the economic intuition of our framework. In contrast, exposure to non-implementable anomalies is associated with lower management fees, suggesting that investors recognize the distinction between implementable and non-implementable anomalies for a given mutual fund. Overall, these findings indicate that our framework captures salient features of institutional investment behavior and serves as a practical tool for identifying implementable return anomalies across different fund sizes.

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Appendix

A. Distribution of Fund Sizes

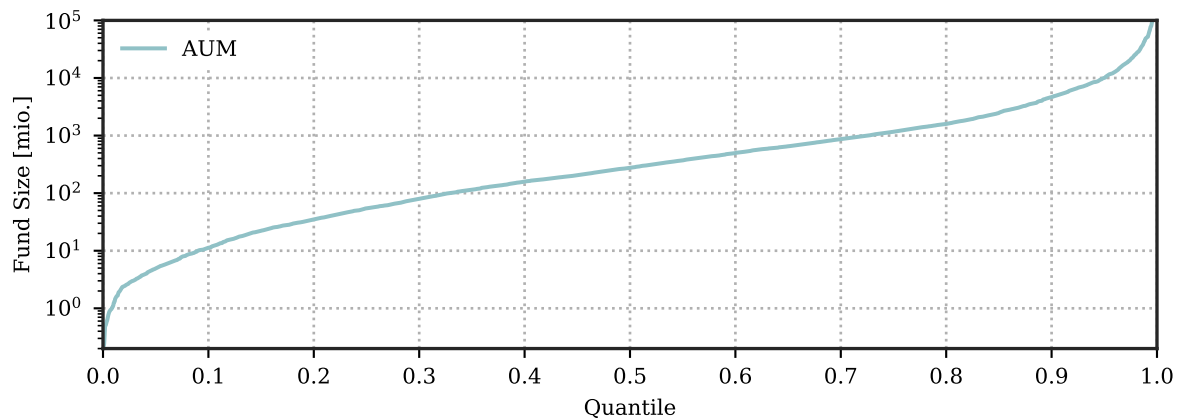


Fig. A. Fund Size Distribution in 2023

The figure shows the fund size distribution of available funds in the CRSP Mutual Fund database for 2023 after filters. We require an equity share based on the percentage of the holdings matched with our JKP sample of at least 50% and a minimum of 10 matched stocks in the portfolio. This leaves us with 4597 different funds in 2023.

B. Summary Statistics

Table B: Mutual Fund Characteristics in Panel Data

The table shows summary statistics of all mutual fund variables in the regression analyses. The variables include the anomaly alignment (anom_align), the monthly return (mret), the annualized management fees (mgmt_fee), the expense ratio (exp_ratio), the turnover ratio (turn_ratio), the log of the fund's total net assets (log_mtna), and its equity share based on the percentage of the holdings matched with our JKP sample (percent_matched).

Variable	N	Mean	Std	Min	Max	25%	75%	IQR
anom_align	64234	0.006	0.097	-0.357	0.570	-0.048	0.066	0.114
mret	64199	0.008	0.076	-0.493	0.685	-0.033	0.048	0.081
mgmt_fee	57503	0.442	0.340	-3.361	1.361	0.250	0.650	0.400
exp_ratio	57473	0.006	0.004	0.000	0.030	0.003	0.009	0.007
turn_ratio	57424	0.856	1.378	0.000	27.390	0.240	0.900	0.660
log_mtna	64234	5.379	1.984	-2.303	10.543	3.970	6.697	2.727
percent_matched	64234	0.879	0.133	0.508	1.000	0.828	0.972	0.144